Linear-Gaussian State Space Models and the Kalman Filter

Group Project Proposal and Individual Extensions

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1 Introduction

A hidden state observed only through noisy measurements can never be known exactly, so Bayesian filtering instead tracks a probability distribution for it, updated as each measurement arrives (Särkkä and Svensson, 2023). This underlies navigation, signal processing, econometrics and weather forecasting, and the linear-Gaussian case is the one setting where it can be solved exactly, by the Kalman filter. We study simulated examples based on vehicle tracking (Särkkä and Svensson, 2023, §4.2) and a drifting regression coefficient such as a hedge ratio (Särkkä and Svensson, 2023, §§3.4–3.5), with a two-dimensional drifting object as a running example throughout.

2 Model and Tasks

2.1 Model and assumptions

The hidden state $x_t \in \mathbb{R}^d$ and the measurement $y_t \in \mathbb{R}^k$ satisfy

$$\begin{aligned} \text{Dynamics:} \quad & x_t = Fx_{t-1} + q_t, && q_t \sim \mathcal{N}(0, Q), \\ \text{Measurement:} \quad & y_t = Hx_t + r_t, && r_t \sim \mathcal{N}(0, R), \\ \text{Prior:} \quad & x_0 \sim \mathcal{N}(m_0, P_0). \end{aligned}$$

A known constant drift u (as in the running example) can be included as $x_t = Fx_{t-1} + u + q_t$, which only changes the prediction mean to $Fm + u$ (Särkkä and Svensson, 2023, §6.4). We assume:

(A1) Markov dynamics: x_t depends on the past only through x_{t-1};
(A2) the measurements are conditionally independent given the current state;
(A3) q_t, r_t and x_0 are mutually independent and Gaussian;
(A4) F, Q, H and R are known (Särkkä and Svensson, 2023, §6.1); parameter estimation is left to the individual extensions.

2.2 Inference tasks

Given the measurements, what do we believe about the hidden state?

- **Filtering:** $p(x_t \mid y_{1:t})$, the state now;
- **Prediction:** $p(x_{t+h} \mid y_{1:t})$, the state in the future;
- **Smoothing:** $p(x_t \mid y_{1:T})$, a past state, using all the data.

2.3 Aim

Derive, prove and implement the two exact algorithms for linear-Gaussian state space models (LGSSMs), and test them in a simulation study:

- the **Kalman filter**, for filtering and prediction;
- the **Rauch–Tung–Striebel (RTS) smoother**, for smoothing.

As the exact solution, the Kalman filter provides the benchmark against which all individual extensions are tested.

3 Background: Linear Algebra, Gaussian Algebra and Bayesian Updating

To keep the report self-contained, we recap the relevant **linear algebra** (block matrices, Schur complements, the matrix inversion lemma; Särkkä and Svensson, 2023, App. A.2) and **Bayesian updating**: with the parameters treated as known, filtering alternates a prediction step (pushing the current belief through the dynamics) and an update step (conditioning on the new measurement by Bayes' rule), tracking the distribution of x_t given $y_{1:t}$ recursively. Three standard facts about Gaussian distributions (Särkkä and Svensson, 2023, App. A.1); (Sanz-Alonso et al., 2023, Ch. 2) imply that every filtering and smoothing distribution remains Gaussian, so it suffices to track a mean and a covariance.

- **Linear propagation:** if $x \sim \mathcal{N}(m, P)$ and $x' \mid x \sim \mathcal{N}(Ax, B)$, then $x' \sim \mathcal{N}(Am,\, APA^\top + B)$, by the laws of total expectation and total variance. This yields the prediction step.
- **Joint Gaussianity:** if $x \sim \mathcal{N}(m, P)$ and $y \mid x \sim \mathcal{N}(Hx, R)$, then (x, y) is jointly Gaussian with $\mathrm{Cov}(x, y) = PH^\top$ and $\mathrm{Cov}(y) = HPH^\top + R$.
- **Conditioning:** if (x, y) is jointly Gaussian with means m_x, m_y and covariance blocks P_{xx}, P_{xy}, P_{yy}, then $x \mid y \sim \mathcal{N}(m_{x|y}, P_{x|y})$ with

$$m_{x|y} = m_x + P_{xy}P_{yy}^{-1}(y - m_y), \qquad P_{x|y} = P_{xx} - P_{xy}P_{yy}^{-1}P_{yx}.$$

  This yields the update step.

4 Methods

4.1 Kalman filter

A forward, online recursion on (m_t, P_t) (Kalman, 1960). For $t = 1, \ldots, T$:

$$\begin{aligned} \text{Prediction:} \quad & m'_t = Fm_{t-1}, \qquad P'_t = FP_{t-1}F^\top + Q, \\ \text{Update:} \quad & K_t = P'_t H^\top \left(HP'_t H^\top + R\right)^{-1}, \\ & m_t = m'_t + K_t\,(y_t - Hm'_t), \qquad P_t = (I - K_t H)\,P'_t. \end{aligned}$$

4.2 RTS smoother

A backward pass that incorporates future data (Rauch et al., 1965). Starting from the final filter output, $m^s_T = m_T$ and $P^s_T = P_T$, for $t = T - 1, \ldots, 1$:

$$\begin{aligned} \text{Smoother gain:} \quad & G_t = P_t F^\top (P'_{t+1})^{-1}, \\ \text{Mean:} \quad & m^s_t = m_t + G_t\,(m^s_{t+1} - m'_{t+1}), \\ \text{Covariance:} \quad & P^s_t = P_t + G_t\,(P^s_{t+1} - P'_{t+1})\,G_t^\top. \end{aligned}$$

Here m_t, P_t are the filter outputs, m'_{t+1}, P'_{t+1} the filter's predictions, and the superscript s marks smoothed quantities.

4.3 Theoretical support

- **Derivations with justification** of both recursions from the facts in Section 3, and **correctness:** exactness, batch equivalence, optimality (minimum mean-squared error), stability, and exact likelihood evaluation via the prediction-error decomposition (Särkkä and Svensson, 2023, Ch. 6, 12, 16); (Sanz-Alonso et al., 2023, Ch. 7–8).
- **Computational cost and scaling** of the basic Kalman filter: $O(d^3 + k^3)$ per step, versus $O\big((Td)^3\big)$ for the batch solution.
- **Worked examples:** the scalar case $\dim x = \dim y = 1$, then other simple models.

4.4 Implementation and numerics

A shared, tested Python codebase, reused by every extension, will implement any LGSSM: given $(F, Q, H, R, m_0, P_0, y_{0:T})$, it checks basic constraints and returns the filtering and smoothing distributions and the likelihood, with Joseph-form and square-root variants for numerical stability. Each unit test gives numerical evidence for the theoretical support above.

(1) **Exactness:** agreement with the batch posterior to machine precision on small problems.
(2) **Accuracy:** RMSE of filter and smoother against the true simulated state, on the running and tracking examples; visualisation of filtering versus smoothing uncertainty.
(3) **Calibration:** coverage of 95% credible intervals and the normalised estimation error squared (NEES) (Bar-Shalom et al., 2001).
(4) **Scalability and robustness:** run time as $d = \dim x$ and $k = \dim y$ grow, compared with the predicted scaling; missing observations and time-varying models.

5 Individual Extensions

For each extension, we will identify a concrete dataset early on.

Aseem Kulkarni: Pairs Trading with State Space Models

- Explore pairs trading by modelling the relationship (ratio) between two closely related stocks as an LGSSM, and test the model on simulated data.

- Backtest the resulting strategy on real market data, compare it against simple benchmark trading strategies, and analyse the results.
- Potential to extend to a high-dimensional setting, e.g. many pairs at once.

Victor Sze: Interacting Particle Systems

(1) **Model:** N interacting agents, each modelled individually, $x_t^{(i)} = A\, x_{t-1}^{(i)} + B\, \bar{x}_{t-1} + q_t^{(i)}$, where $\bar{x}_t$ is the average over all agents and B measures how strongly each agent is influenced by the group. Stacking all agents gives an LGSSM of dimension Nd.
(2) **Known parameters first:** filter the full system with the Kalman filter.
(3) **Decoupled Kalman filter:** changing coordinates to the group average and orthogonal deviations splits the filter into independent d-dimensional filters, giving the same exact answer at a cost growing linearly rather than cubically in N.
(4) **Application:** pick an application with a concrete dataset of interacting agents.
(5) **Possible directions (to be discussed):** estimating the interaction parameters, e.g. by MCMC using the Kalman likelihood; comparing approximate filters (EnKF, particle filter) with the exact filter for large N.

Nigel Prem Lim: Tracking Footballers' Market Values

- A player's true market value cannot be observed directly; we only see indirect signals, such as on-pitch performance, published valuations (e.g. Transfermarkt) and occasional actual transfer fees.
- Treat the true value as a hidden state that evolves over time, and use the Kalman filter to estimate it from these noisy, irregularly observed signals.
- Identify usable data and, given the data, come up with a credible likelihood model, e.g. p(on-pitch performance | value).

Lozko Pilev: Parameter Estimation

- Parameter estimation in general: relax the assumption of known F, Q, H, R and study the **EM algorithm**, which alternates between estimating the hidden states $x_{0:T}$ (via the RTS smoother) and the parameters θ.
- Filtering under model misspecification, linked to filter stability: compare the effect of different choices of Q and R (e.g. for a time-varying hedge ratio) on the accuracy and uncertainty of the state estimate, and ask whether $\|F - F_*\| \leq \varepsilon$ implies $\|m_t - m_{t,*}\| \leq \varepsilon'$.

References

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